

NITHISH KUMAR V

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PROFESSIONAL SUMMARY

MCA candidate (CGPA 8.96) with hands-on experience designing and deploying production data systems using AI-assisted development. Built the Faculty Workload Management System — a 22-table PostgreSQL platform serving 50+ faculty — and an AI-powered family finance PWA with Google Gemini integration. Targeting Data Analyst and Business Analyst roles at technology and consulting organisations.

TECHNICAL SKILLS

SQL & Databases: PostgreSQL, MySQL | JOINS, GROUP BY, HAVING, COALESCE, Aggregations, Subqueries | Schema Design, Data Modelling

Languages & Frameworks: Python (FastAPI, SQLAlchemy), JavaScript, TypeScript, React 18 | REST API Development | Pydantic, Zustand

Analytics & Reporting: Workload Deviation Analysis, KPI Design, Budget Utilisation Metrics | Excel Multi-Sheet Reporting | Power BI, Recharts

AI Integration: Google Gemini API (chatbot, NLP parsing), AI-assisted system development | Prompt Engineering

Tools & BA: Git, Postman, Supabase, Netlify, Vercel, Google OAuth 2.0 | Requirements Documentation, Stakeholder Mapping

PROJECTS

Faculty Workload Management System (FWMS) | *Python · FastAPI · PostgreSQL · React · TypeScript · Google OAuth*

- Designed and deployed a 22-table PostgreSQL system automating subject allocation for 50+ faculty across 4 academic programs, 150+ courses, and 19,200+ offerings — delivered to department for institutional adoption
- Built analytics layer computing per-faculty workload deviation against an 18-hour TCH norm, classifying allocations as **Overloaded / Balanced / Underloaded**; automated 3-sheet Excel exports (15 metrics per faculty) and PDF summaries
- Wrote SQL aggregation queries using LEFT JOINS, COALESCE, GROUP BY, and subqueries to compute per-faculty TCH totals, deviation hours, and department KPIs (allocation success rate, overload %)
- Engineered a 5-pass progressive allocation algorithm with 9 business rules; implemented 4-tier RBAC (Faculty / Coordinator / HOD / Admin) across 30+ REST API endpoints with full audit logging

Family Finance Tracker | *React 18 · FastAPI · PostgreSQL · Google Gemini AI · PWA · Recharts · Supabase*

- Built and deployed an AI-powered family finance PWA tracking pension income, multi-category expenses, monthly budgets, and a multi-asset portfolio (Fixed Deposits, Mutual Funds, LIC, Chit Funds, Gold) across 3 users — live at family-finance-tracker-pearl.vercel.app
- Integrated Google Gemini AI for SMS-based natural language expense parsing, an AI financial chatbot (Ask Amma), and automated monthly reports with AI-generated insights and one-click Claude export
- Designed analytics computing income-vs-expense deficit, per-category budget utilisation, and portfolio net worth across 5 asset classes — visualised with Recharts dashboards, offline-capable via PWA service worker

VaxVault — Vaccination Reminder System | *Node.js · Express.js · MongoDB · React · Tailwind CSS*

- Contributed to a full-stack vaccination reminder app with multi-profile scheduling, automated email notifications via 3-tier cron jobs, and a KPI dashboard tracking total profiles, pending, completed, and overdue vaccinations
- Modelled data validation rules for scheduling conflict detection and duplicate booking prevention; identified 8+ edge cases across booking, notification, and record management workflows

EDUCATION

MCA — Hindustan Institute of Technology & Science, Chennai | CGPA: 8.96 / 10 | Expected: June 2026

BCA — Hindustan College of Arts & Science, Chennai | CGPA: 7.96 / 10 | April 2024

HSC — AEHSS-2, Chennai (State Board) | 85.86% | 2021

SSC — AECS-2, Chennai (CBSE) | 80.40% | 2019

CERTIFICATIONS

- Cloud Computing Fundamentals — NPTEL (2025)